SOUMIKA DAS

+91 8617233878 ⋄ Kolkata, India
dassoumika11@gmail.com ⋄ linkedin.com/in/soumikadas ⋄ github.com/Soumika-11

EXPERIENCE

Software Engineer Intern Oct 2025 – Present
PepHub (Service-Based) Remote

- Engineered 3 full-stack platforms (Next.js/Node.js portal with Stripe, ASP.NET Core FinTech API, Vite botanical tool) achieving 99.8% payment success, 55% less dependencies, 80% faster loads.
- Delivered 70% faster manual processing, 3x deployment acceleration, 68% bug reduction, 60% workflow improvement via modular architecture.
- Managed 3 production deployments with PM2/SSH maintaining 99.9% uptime while reducing deployment time from 45min to 8min.

Software Engineer Intern (Backend Systems) Jan 2025 - Jun 2025
RevMaxx (HealthTech) Onsite

- Developed HIPAA-compliant medical transcription backend with SNOMED CT/ICD-10 reducing coding errors 45%, improving compliance 78→ 96%, accelerating claims 40%.
- Built AI microservice for medical entity extraction using NLP achieving 93% accuracy across 50K+ clinical notes, cutting manual review 65%.
- Implemented error handling with retry logic/circuit breakers reducing failures 92% (12/week→ ¡1) & achieving 99.7% uptime.

Software Engineer Intern (Backend & Data) Jun 2025 - Sep 2025
Cardiacrhythm (HealthTech) Remote

- Built WebSocket real-time ECG streaming infrastructure supporting 200+ concurrent streams, reducing latency 90% (1.8s→ 180ms) with ¡1% packet loss.
- Optimized large JSON payloads via compression/chunking reducing size 75% (8MB→ 2MB), cutting bandwidth 60%, improving responsiveness 4x.
- Implemented error recovery with auto-reconnection/session restoration increasing persistence 82→ 97%, eliminating data loss for 15K+ daily sessions.

Research Engineer Intern (Developer) Dec 2024 - Present
IEDC, Dept. of CSE, UEM Kolkata Hybrid

- Developed brain tumor classification via CNN-LSTM achieving 94.3% accuracy on 3,200+ MRI scans; published AISC Conference paper with 18% improvement.
- Architected 2 HPC systems: "Satyameba" cluster (Ansible automation, 8hrs→ 22min setup, 3 news features) & underwater submarine communication (85% delivery, 40% latency cut).
- Engineered real-time YOLO/OpenCV face detection at 96% accuracy, 45 FPS across 8 streams; enabled 5x faster ML training infrastructure.

SKILLS

- **Languages:** C#, Python, TypeScript/JavaScript, SQL
- **Backend:** ASP.NET Core, Node.js, FastAPI, REST APIs, WebSocket, Microservices
- **Frontend:** Next.js, React, Vite
- **Databases:** PostgreSQL, MS SQL Server, Redis
- **DevOps:** AWS EC2, PM2, Docker, Ansible, Linux, Nginx
- **ML/AI:** CNN, LSTM, YOLO, TensorFlow, PyTorch, OpenCV, NLP
- **Healthcare:** HIPAA, SNOMED CT, ICD-10

PAPER & PUBLICATIONS

- **Brain Tumor Detection using CNN-LSTM**, AISC Conference (18% accuracy improvement via ensemble learning)

EDUCATION

Bachelor of Technology in Computer Science and Engineering Expected 2027
(IoT, Cybersecurity and Blockchain Technology) GPA:8.02
Institute of Engineering and Management, Newtown

PROJECTS

- **Real-Time Video Proctoring:** Engineered exam monitoring with WebSocket, OpenCV & MediaPipe achieving 91% violation detection accuracy.
- **AI Resume Parser:** Built NLP parser extracting structured JSON from PDF/DOCX with 94% accuracy; reduced screening time by 8hrs/week & cut time-to-hire by 40%.
- **Diaspora Job Marketplace:** Launched production job platform with JWT auth, OAuth & payment integration; achieved 99.7% uptime.